

Play Black Piano Keys

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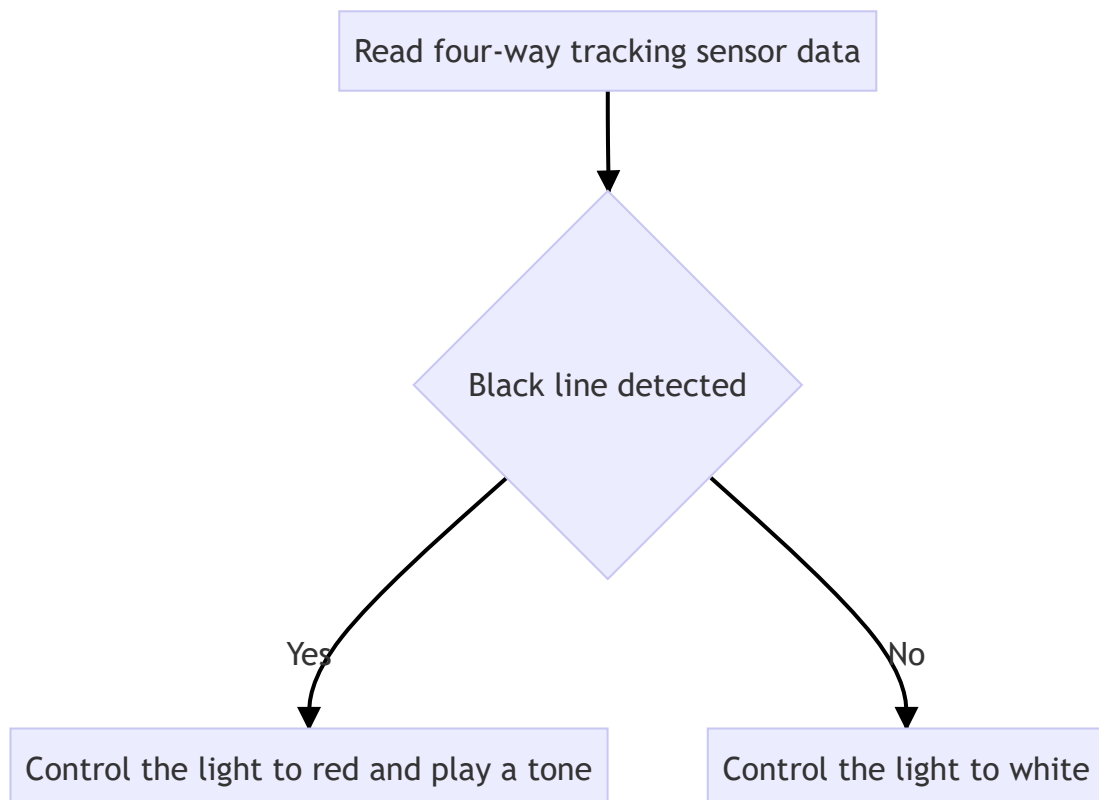
1. Experiment Purpose

In this section, we will learn to use the tracking sensor to detect black lines and make the car "play" notes. When the car detects a black line, it will emit sounds of different pitches, as if playing a piano.

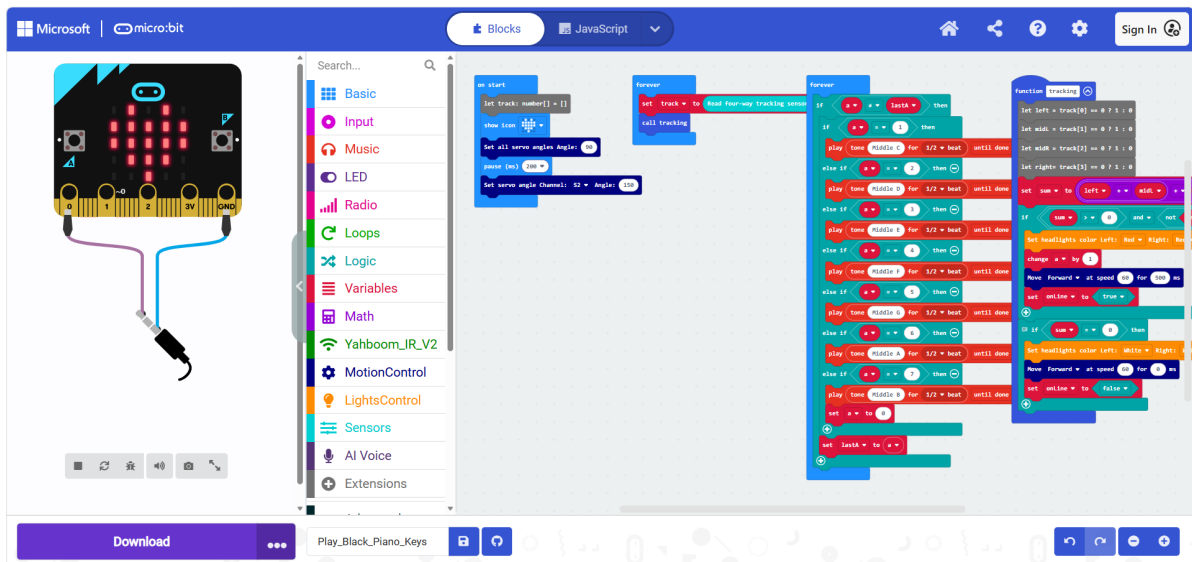
2. Hardware Requirements

Make sure the battery is fully charged and the four-way tracking sensor has been adjusted to a suitable sensitivity with the potentiometer.

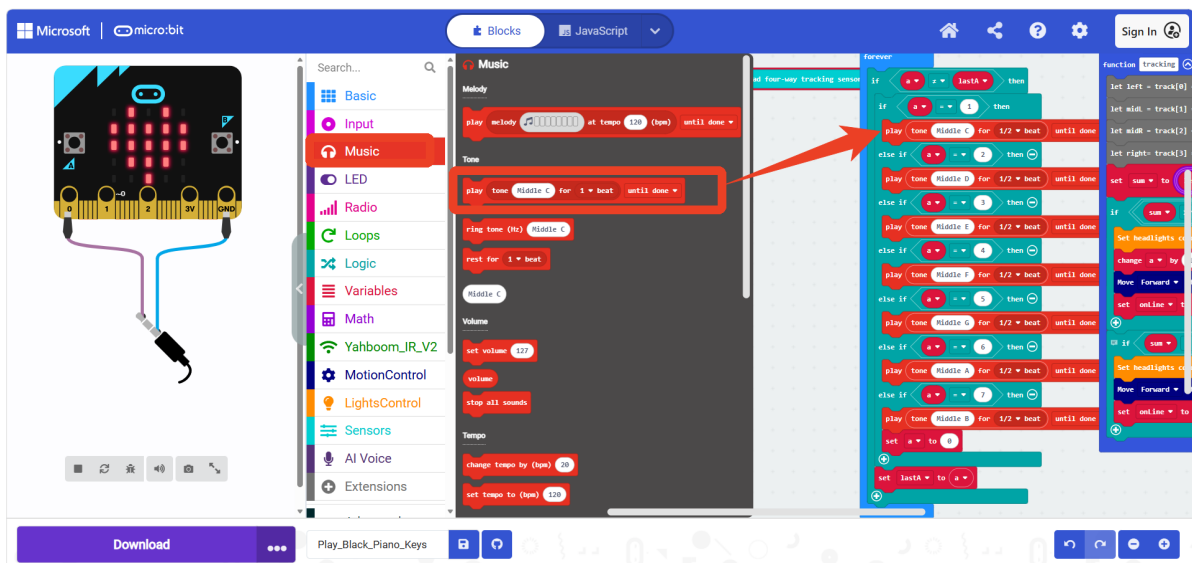
3. Implementation Principle



The program is as follows.



The play tone block is located in the **Music** category.



Other blocks, such as motor control and servo control, are located in the **Lastbit Motion Control** category, which has already been covered in previous tutorials, so we will not repeat it here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `3.Play_Black_Piano_Keys.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

The car moves forward. **Due to mechanical errors in the motors and track slippage during movement, the car may not drive in a straight line even if the left and right motors are set to the same speed.**

When the four-way tracking sensor detects a black line, it lights up red and plays different tones.

When none of the sensors detects black, the light turns white.

